[INCH–POUND]

A–A–59789/22A
22 October 2016
SUPERSEDING
A–A–59789/22
22 October 2011

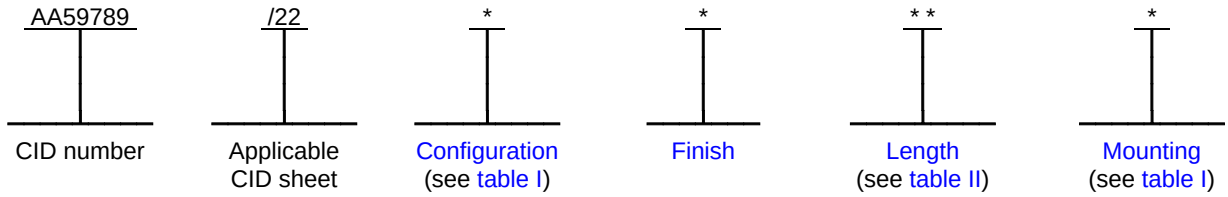
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 5 PIECE, FOR COLD PLATE APPLICATIONS, .225 X .260 INCH BODY SIZE, WEDGE MOUNTING BODY, WITH TWO OR NO MOUNTING HOLES AND VISUAL LOCK INDICATION, SCREW ACTUATED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59789.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes herein).



Example: AA59789/22CR65S is the PIN for a clear chemical film finished, 6.3 inch (160 mm) long card holder. The card holder also features two tapped mounting holes for use with 2–56 UNC 2B fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in its installed positions and prevent loosening or movement as a result of shock and vibration. It shall also provide a thermal transfer path from the circuit card assembly to the cold plate.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and meet the general requirements specified in CID A–A–59789. The card holders shall have a visual indicator to show when the assembly is in its relaxed (unlocked) and expanded (locked) state.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59789.

Actuating screw hex drive socket. The dimension for hex drive socket shall be .094 inch (2.38 mm) across flats for mounting options "N", "S", "R", "P", "U", and "T".

Cold plate slot width. The recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59789).

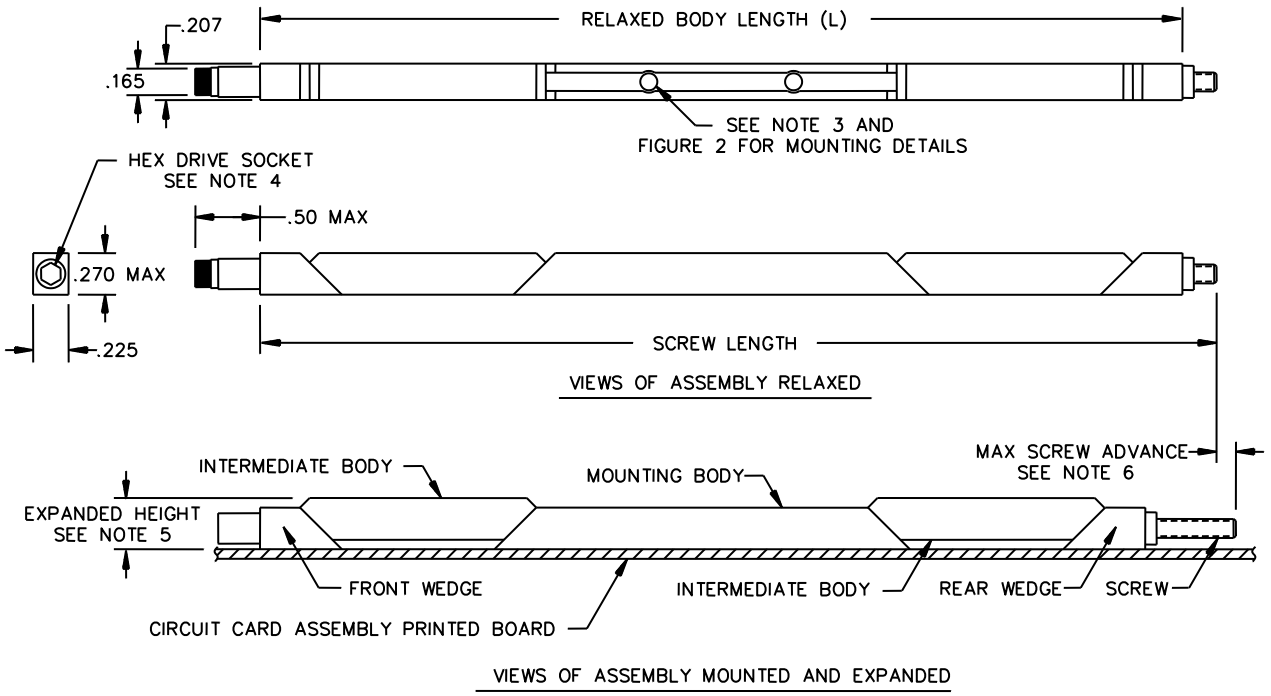
Installation torque. The recommended nominal installation torque is as follows: 6 inch-pounds (68 N-cm) for assemblies of configuration "A", "C", "F", or "W" and 7.2 inch-pounds (81 N-cm) for assemblies of configurations "D", "J", "L", or "N".

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Configuration. The configuration of a card holder shall be as specified in table I. The details of a particular configuration consist of those on figures 1, 2 and 3, and may include those on figures 4 or 5.

TABLE I. Configuration.

Configuration	Mounting body style (see figure 2)	Mounting options	Applicable figures	Hardware options
A	A	P, S, R, T	1, 2 and 3	No hardware options added
C	C	N, S, R, T		
F	A	P, S, R, T	1, 2, 3 and 4	Lockwasher and flat washer
W	C	N, S, R, T		
N	A	P, S, R, T	1, 2, 3 and 5	Screw self-lock element
L	C	N, S, R, T		
J	A	P, S, R, T	1, 2, 3, 4 and 5	Lockwasher, flat washer, and screw self-lock element
D	C	N, S, R, T		



Inches	mm	Inches	mm	Inches	mm
.125	3.18	.225	5.72	.30	7.6
.140	3.56	.270	6.86	.900	22.86

FIGURE 1. Relaxed and expanded dimensions (two mounting hole version shown).

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NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Screw and alignment spring not shown. See table II and figure 2 for hole spacing requirements.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm).
5. Mounting body style "A" is capable of safely expanding to .330 inch (8.38 mm) and mounting body style "C" is capable of expanding to .325 inch (8.26 mm) minimum.
6. Maximum screw advance is .12 inch (3.0 mm) for mounting body style "A" and .14 inch (3.6 mm) for mounting body style "C".

FIGURE 1. Relaxed and expanded dimensions – Continued.

Visual lock indicator (see figure 2). Card holders shall have a visual indicator to show when the card holder is in its relaxed (unlocked) state. When the card holder is in the unlocked (relaxed) state, the end of the actuating screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

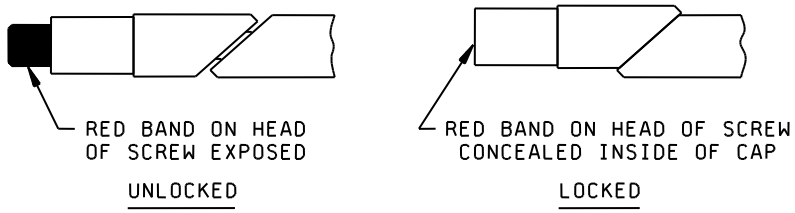


FIGURE 2. Visual lock indicator.

Finish. The finish designator shall be as specified in A–A–59789. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "E" (electroless nickel), "H" (hard black anodize), or "R" (clear chemical film).

Length, expanded and relaxed dimensions. The length designator shall be as specified in A–A–59789. The lengths available for this CID specification sheet are listed in table II. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

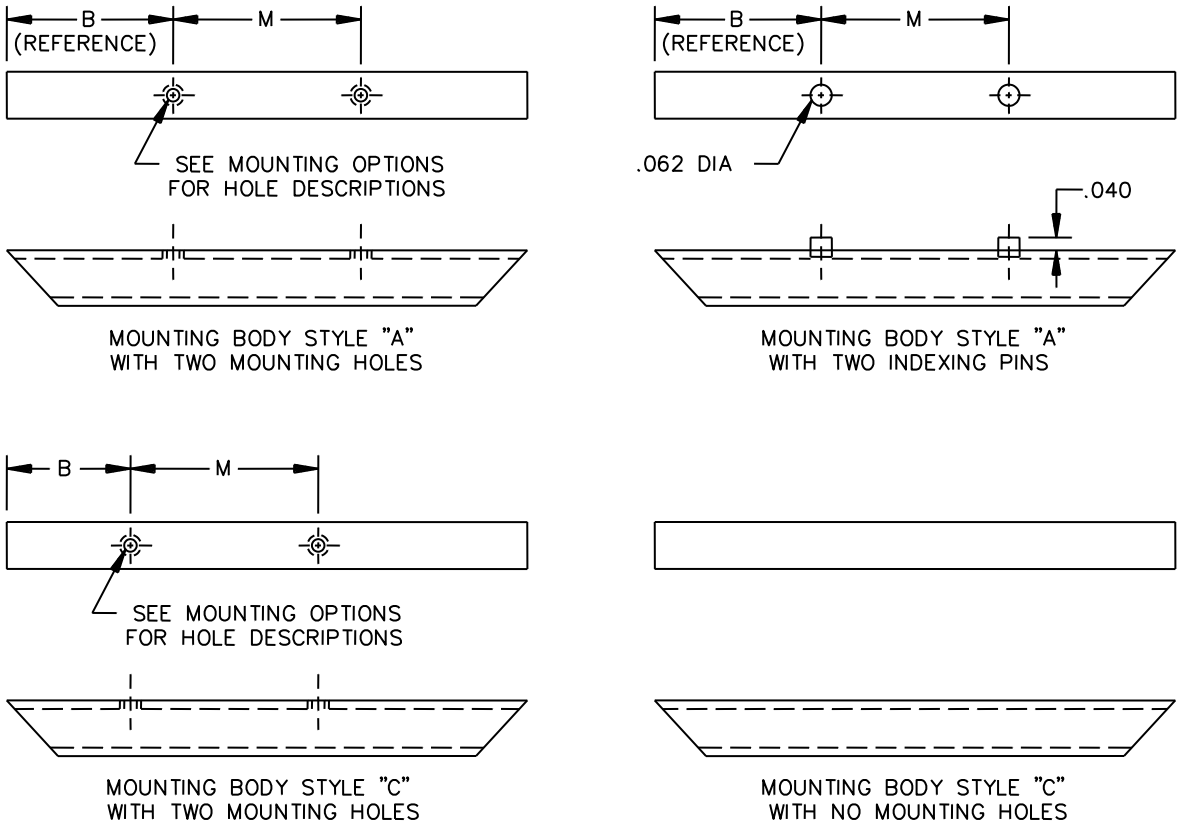
TABLE II. Additional assembly dimensions (see figures 1 and 2). 1/

PIN length designator	Dimension "L" ±.02 (0.5)	Mounting body style	Dimension "A" ±.01 (0.3)	Dimension "B" ±.02 (0.5)	Dimension "M" 2/
50	4.8 (121.9)	A	1.50 (38.1)	.30 (7.6)	.900
		C	1.60 (40.6)	.35 (8.9)	.900
55	5.3 (134.6)	A	2.00 (50.8)	.55 (14.0)	.900
		C	1.60 (40.6)	.35 (8.9)	.900
60	5.8 (147.3)	A	2.50 (63.5)	.80 (20.3)	.900
		C	1.60 (40.6)	.35 (8.9)	.900
65	6.3 (160.0)	A	3.00 (71.2)	1.05 (26.7)	.900
		C	1.60 (40.6)	.35 (8.9)	.900

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for general information only.
2/ Tolerance for the hole spacing is ±.005 inch (0.51 mm).

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Mounting body style (see figure 3 and table II). Two mounting body styles are available on this CID specification sheet, style A or style C. The primary difference in the two mounting body styles are the location of mounting holes on the mounting body as depicted on figure 2. Mounting body style A always has two locations for fasteners centered on the mounting body that are .900 inch (22.86 mm) apart. Mounting body style C can have two or no holes on the mounting body.



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. See table II for dimensions B, E, and M.

FIGURE 3. Mounting body style details.

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Mounting. The mounting designators shall be as specified in A–A–59789. The mounting options available for this CID specification sheet are as follows: "N" (no mounting holes; mounting body style C only), "P" (Ø.062 x .040 index pins; mounting body style A only), "R" (rivet mount holes with counterbore and countersink), "S" (tapped 2–56 holes), or "T" (tapped 0–80 holes). See figure 1 for mounting hole spacing requirements. Card holders using mounting option "P" and "R" may be shipped unassembled.

Rivet mount holes. The holes used for rivet mounting shall be .066/.073 inch (1.68/1.85 mm) diameter, countersunk 100 degrees by .140 inch (3.56 mm) diameter through holes with a counterbore of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

Rivets. This card holder uses rivet style A as specified in A–A–59789 when rivet mounting is used.

Hardware options. Card holders can have the following hardware options. No added hardware options, lock washer and flat washer (see figure 4), screw self locking element (see figure 5), or a combination of both lock washer and flat washer and screw self locking element. See table I for the correct PIN configuration identifier.

No added hardware. The base configuration of the card holder is as depicted by figures 1, 2 and 3. Card holders requiring no added hardware options shall include configuration identifier "A" or "C" in the PIN (see classification, table I and notes section herein).

Lockwasher and flat washer (see figure 4). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include configuration identifier "F" or "W" in the PIN (see classification, table I and notes sections herein).

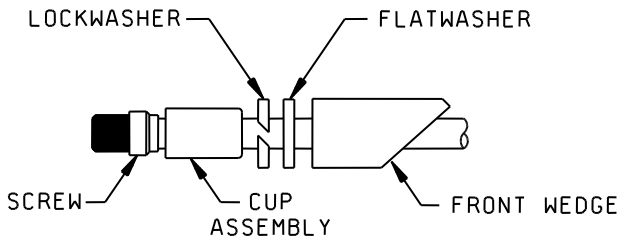


FIGURE 4. Lockwasher and flat washer details.

Screw self-locking element (see figure 5). The use of a screw self-locking element will provide prevailing torque for increased resistance to loosening of the card holder assembly from shock vibration. The screw self-locking element shall be in accordance with A–A–59789. Card holders requiring a screw self-locking element shall include configuration identifier "L" or "N" in the PIN (see classification, table I and notes sections herein).

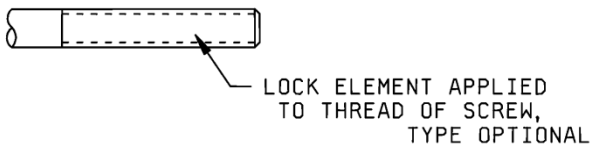


FIGURE 5. Screw self-locking element details.

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Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include configuration identifier "D" or "J" in the PIN (see classification, table I and notes sections herein).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59789 – Holder, Electrical Card, Wedge Retainers, 5 Piece, For Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

Ordering data. Ordering data shall be as specified in A–A–59789.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
5BG68	American Circuit Card Retainers, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806–1231	Telephone: (714) 738–6194 Facsimile: (714) 446–0119 E-mail: sales@accrmfg.com URL: http//www.accrmfg.com
61081	Pentair Technical Products (formerly Birtcher) 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 E-mail: schroff.us@pentair.com URL: http://schroff.pentair.com

Part number supersession data. These CID specification sheet PINs supersede the following manufacturer's part numbers in table III and on DESC Drawing 89064 (table IV) as shown. The CID PINs listed in table III are only for length designator "50". See tables V and VI for CID PIN construction using other available lengths for this CID specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

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TABLE III. Commercial part number supersession data.

PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AB50S	40–5VI–10–B	CB50S	G5225VBA–4.80T2
AB50R	40–5VI–10–B–R	CB50R	G5225VBA–4.80H
AB50T	40–5VI–10–B–T	CB50T	G5225VBA–4.80T0
AB50P	40–5VI–10–B–P	CB50U	G5225VBA–4.80TM2
FB50S	40–5VI–10–B–LF	WB50S	G5225VWBA–4.80T2
FB50R	40–5VI–10–B–LF–R	WB50R	G5225VWBA–4.80H
FB50T	40–5VI–10–B–LF–T	WB50T	G5225VWBA–4.80T0
FB50P	40–5VI–10–B–LF–P	WB50U	G5225VWBA–4.80TM2
NB50S	40–5VI–10–B–L	LB50S	G5225VBA–4.80T2L
NB50R	40–5VI–10–B–R–L	LB50R	G5225VBA–4.80HL
NB50T	40–5VI–10–B–T–L	LB50T	G5225VBA–4.80T0L
NB50P	40–5VI–10–B–P–L	LB50U	G5225VBA–4.80TM2L
JB50S	40–5VI–10–B–LF–L	DB50S	G5225VWBA–4.80T2L
JB50R	40–5VI–10–B–LF–R–L	DB50R	G5225VWBA–4.80HL
JB50T	40–5VI–10–B–LF–T–L	DB50T	G5225VWBA–4.80T0L
JB50P	40–5VI–10–B–LF–P–L	DB50U	G5225VWBA–4.80TM2L
AC50S	40–5VI–10	CC50S	G5225VCG–4.80T2
AC50R	40–5VI–10–R	CC50R	G5225VCG–4.80H
AC50T	40–5VI–10–T	CC50T	G5225VCG–4.80T0
AC50P	40–5VI–10–P	CC50U	G5225VCG–4.80TM2
FC50S	40–5VI–10–LF	WC50S	G5225VWCG–4.80T2
FC50R	40–5VI–10–LF–R	WC50R	G5225VWCG–4.80H
FC50T	40–5VI–10–LF–T	WC50T	G5225VWCG–4.80T0
FC50P	40–5VI–10–LF–P	WC50U	G5225VWCG–4.80TM2
NC50S	40–5VI–10–L	LC50S	G5225VCG–4.80T2L
NC50R	40–5VI–10–R–L	LC50R	G5225VCG–4.80HL
NC50T	40–5VI–10–T–L	LC50T	G5225VCG–4.80T0L
NC50P	40–5VI–10–P–L	LC50U	G5225VCG–4.80TM2L

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
JC50S	40–5VI–10–LF–L	DC50S	G5225VWCG–4.80T2L
JC50R	40–5VI–10–LF–R–L	DC50R	G5225VWCG–4.80HL
JC50T	40–5VI–10–LF–T–L	DC50T	G5225VWCG–4.80T0L
JC50P	40–5VI–10–LF–P–L	DC50U	G5225VWCG–4.80TM2L
AP50S	40–5VI–10–N	CE50S	G5225VEN–4.80T2
AP50R	40–5VI–10–N–R	CE50R	G5225VEN–4.80H
AP50T	40–5VI–10–N–T	CE50T	G5225VEN–4.80T0
AP50P	40–5VI–10–N–P	CE50U	G5225VEN–4.80TM2
FP50S	40–5VI–10–N–LF	WE50S	G5225VWEN–4.80T2
FP50R	40–5VI–10–N–LF–R	WE50R	G5225VWEN–4.80H
FP50T	40–5VI–10–N–LF–T	WE50T	G5225VWEN–4.80T0
FP50P	40–5VI–10–N–LF–P	WE50U	G5225VWEN–4.80TM2
NP50S	40–5VI–10–N–L	LE50S	G5225VEN–4.80T2L
NP50R	40–5VI–10–N–R–L	LE50R	G5225VEN–4.80HL
NP50T	40–5VI–10–N–T–L	LE50T	G5225VEN–4.80T0L
NP50P	40–5VI–10–N–P–L	LE50U	G5225VEN–4.80TM2L
JP50S	40–5VI–10–N–LF–L	DE50S	G5225VWEN–4.80T2L
JP50R	40–5VI–10–N–LFRL	DE50R	G5225VWEN–4.80HL
JP50T	40–5VI–10–N–LFTL	DE50T	G5225VWEN–4.80T0L
JP50P	40–5VI–10–N–LFPL	DE50U	G5225VWEN–4.80TM2L
AH50S	40–5VI–10–B3	CH50S	G5225VBH–4.80T2
AH50R	40–5VI–10–B3–R	CH50R	G5225VBH–4.80H
AH50T	40–5VI–10–B3–T	CH50T	G5225VBH–4.80T0
AH50P	40–5VI–10–B3–P	CH50U	G5225VBH–4.80TM2
FH50S	40–5VI–10–B3–LF	WH50S	G5225VWBH–4.80T2
FH50R	40–5VI–10–B3–LFR	WH50R	G5225VWBH–4.80H
FH50T	40–5VI–10–B3–LFT	WH50T	G5225VWBH–4.80T0
FH50P	40–5VI–10–B3–LFP	WH50U	G5225VWBH–4.80TM2

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/22	Vendor similar designator or or type part number 1/ 2/ 4/ CAGE 5BG68
NH50S	40–5VI–10–B3–L	LH50S	G5225VBH–4.80T2L
NH50R	40–5VI–10–B3–R–L	LH50R	G5225VBH–4.80HL
NH50T	40–5VI–10–B3–T–L	LH50T	G5225VBH–4.80T0L
NH50P	40–5VI–10–B3–P–L	LH50U	G5225VBH–4.80TM2L
JH50S	40–5VI–10–B3–LFL	DH50S	G5225VWBH–4.80T2L
JH50R	40–5VI–10–B3LFRL	DH50R	G5225VWBH–4.80HL
JH50T	40–5VI–10–B3LFTL	DH50T	G5225VWBH–4.80T0L
JH50P	40–5VI–10–B3LFPL	DH50U	G5225VWBH–4.80TM2L

1/ The CID PINs listed are only for length designator "50".
2/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.
3/ Mounting body style A.
4/ Mounting body style C.

TABLE IV. Supersession information.

Superseded (old) PIN in accordance with DESC Drawing 89064	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DESC Drawing 89064	Superseding (new) CID PIN
89064–50ANV	AA59789/22AC50N	89064–50ANWV	AA59789/22FC50N
89064–50ARV	AA59789/22AC50R	89064–50ARWV	AA59789/22FC50R
89064–50ASV	AA59789/22AC50S	89064–50ASWV	AA59789/22FC50S
89064–50ATV	AA59789/22AC50T	89064–50ATWV	AA59789/22FC50T
89064–60ANV	AA59789/22AC60N	89064–60ANWV	AA59789/22FC60N
89064–60ARV	AA59789/22AC60R	89064–60ARWV	AA59789/22FC60R
89064–60ASV	AA59789/22AC60S	89064–60ASWV	AA59789/22FC60S
89064–60ATV	AA59789/22AC60T	89064–60ATWV	AA59789/22FC60T
89064–65ANV	AA59789/21AC65N	89064–65ANWV	AA59789/21FC65N
89064–65ARV	AA59789/21AC65R	89064–65ARWV	AA59789/21FC65R
89064–65ASV	AA59789/21AC65S	89064–65ASWV	AA59789/21FC65S
89064–65ATV	AA59789/21AC65T	89064–65ATWV	AA59789/21FC65T

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TABLE IV. Supersession information – Continued.

Superseded (old) PIN in accordance with DESC Drawing 89064	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DESC Drawing 89064	Superseding (new) CID PIN
89064–50ENV	AA59789/22CC50N	89064–50ENWV	AA59789/22WC50N
89064–50ERV	AA59789/22CC50R	89064–50ERWV	AA59789/22WC50R
89064–50ESV	AA59789/22CC50S	89064–50ESWV	AA59789/22WC50S
89064–50ETV	AA59789/22CC50T	89064–50ETWV	AA59789/22WC50T
89064–60ENV	AA59789/22CC60N	89064–60ENWV	AA59789/22WC60N
89064–60ERV	AA59789/22CC60R	89064–60ERWV	AA59789/22WC60R
89064–60ESV	AA59789/22CC60S	89064–60ESWV	AA59789/22WC60S
89064–60ETV	AA59789/22CC60T	89064–60ETWV	AA59789/22WC60T
89064–65ENV	AA59789/22CC65N	89064–65ENWV	AA59789/22WC65N
89064–65ERV	AA59789/22CC65R	89064–65ERWV	AA59789/22WC65R
89064–65ESV	AA59789/22CC65S	89064–65ESWV	AA59789/22WC65S
89064–65ETV	AA59789/22CC65T	89064–65ETWV	AA59789/22WC65T

PIN length examples. The CID PINs listed in table IV are for all available standard card holder lengths for this specification sheet. However, only one specific finish, mounting and configuration are listed (see PIN example for a break-down of the codes).

TABLE V. Example of PIN with available length designators (two hole mounting bodies).

PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AB50S	40–5VI–10–B	CB50S	G5225VBA–4.80T2
AB55S	40–5VI–11–B	CB55S	G5225VBA–5.30T2
AB60S	40–5VI–12–B	CB60S	G5225VBA–5.80T2
AB65S	40–5VI–13–B	CB65S	G5225VBA–6.30T2

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.
2/ Other lengths are available on request.
3/ Mounting body style A.
4/ Mounting body style C.

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TABLE VI. Examples of PINs with available length dimensions (no mounting holes in bodies).

PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/22	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AB50P	40–5VI–10–B–P	CB50N	G5225VBA–4.80
AB55P	40–5VI–11–B–P	CB55N	G5225VBA–5.30
AB60P	40–5VI–12–B–P	CB60N	G5225VBA–5.80
AB65P	40–5VI–13–B–P	CB65N	G5225VBA–6.30

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.
2/ Other lengths are available on request.
3/ Mounting body style A with indexing pins in the mounting body.
4/ Mounting body style C with no holes in the mounting body.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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